

BIPIN CHOWDARY

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PROFESSIONAL SUMMARY

AI Engineer and FAU M.S. Artificial Intelligence candidate with experience across electrical systems, cloud engineering, robotics, machine learning, and research software. Builds reproducible systems for marine autonomy, carbon-aware AI, information retrieval, computer vision, and deep learning using measurable baselines, rigorous evaluation, and deployable prototypes.

EDUCATION

- Master of Science in Artificial Intelligence | Florida Atlantic University, Boca Raton, FL | Expected Dec 2026
- Bachelor of Technology (Honors) in AI & Machine Learning | Minor: Robotics & Automation | Quantum University, Roorkee, UK | 2024
- Diploma in Electrical & Electronics Engineering | Minor: Robotics | A.A.N.M. & V.V.R.S.R. Polytechnic, Gudlavalleru, AP | 2021

PROFESSIONAL EXPERIENCE

Mail & Logistics Operations Associate – Exela Technologies, Florida Atlantic University | Boca Raton, FL | Oct 2025 - Present

- Operate digital package-management workflows for parcel intake, recipient verification, automated notifications, exception tracking, and service-desk resolution.

Electrical Engineering Intern - APGENCO | PowerGrid | 6 months

- Increased grid reliability by 20% through optimization of electrical systems in collaboration with engineering teams.
- Contributed to maintenance projects that reduced downtime by 15% over six months.

Web Development Intern - Lets Grow More | Portfolio and Web Design | 2 months

- Improved client online presence by designing five user-friendly websites, resulting in a 45% increase in engagement.
- Enhanced user engagement metrics by 50% through implementation of best practices in web design.

Cloud Computing Intern - Google Cloud | DevOps and Cloud Engineering | 6 months

- Streamlined cloud deployment processes, reducing service delivery time by 40% through optimized infrastructure.
- Enhanced operational efficiency by 20% through collaborative projects focusing on process improvements.

AI & Machine Learning Intern - Xebia | AIML Algorithms and Deep Learning Architectures | 36 months

- Developed advanced AIML models, contributing to two academic research publications.
- Successfully implemented deep learning architectures, achieving 85% model accuracy on real-world applications.

AI & Robotics Automation Intern - RoboLabs AI | Automation and Robotics | 6 months

- Improved robotic function efficiency by 30% through automation solutions integrated with AI technologies.
- Led a team in automation projects, resulting in a 25% reduction in operational time.

SELECTED RESEARCH & TECHNICAL PROJECTS

AquaNavAI - Energy-Aware Marine Navigation | 2026 | [Live Demo](#) | [Read Paper](#)

- Built ASV route planning with A*/Dijkstra baselines, propulsion-energy estimates, offline MapLibre mission visualization, and 30 paired evaluation cases.
- Added NOAA-source adapters, provenance/checksum controls, schemas, CI, and automated Python, frontend, and browser tests.

Carbon-Aware Marine AI | 2025 | [View Case Study](#) | [Read Paper](#)

- Prepared NOAA OISST 2020-2026 data; Ridge forecasting achieved 0.1336 RMSE versus 0.1451 for persistence on the South Florida study subset.
- Tracked experiments with CodeCarbon and estimated ~99.5% emissions savings for the evaluated clean-energy regional schedule.

Automotive Modding Information Retrieval | 2026 | [View Project](#) | [Watch Walkthrough](#)

- Evaluated BM25, dense, and hybrid retrieval on 693 documents, 10 queries, and 200 judgments; BM25 reached P@10 0.46 and nDCG@10 0.3858.

Fuel-efficiency prediction | 2024 | [Live Demo](#)

- Developed a deep-learning workflow integrating anatomical segmentation and dose prediction for automated radiotherapy planning.

Earlier Engineering Projects | 2021-2024 | [Portfolio Projects](#)

- crowd anomaly detection (OpenCV), 4-omni robot navigation, DCGAN image generation, autonomous smart-city automation, and industry-specific web platforms.

TECHNICAL SKILLS

Languages & Data: Python, R, MATLAB, NumPy, Pandas, Matplotlib, Seaborn

AI / ML: PyTorch, TensorFlow, Keras, scikit-learn, deep learning, computer vision, information retrieval, evaluation

Robotics & Embedded: ROS, Gazebo, Arduino, Raspberry Pi, sensors, navigation, control logic

Cloud & Software: Google Cloud, Kubernetes, Docker, Git, GitHub, TypeScript, JavaScript, Streamlit, MapLibre, CI/CD, JSON Schema, CodeCarbon

RESEARCH OUTPUTS

AquaNavAI technical manuscript and software artifact (2026); Carbon-Aware Marine AI graduate report (2026); BM25/dense/hybrid information-retrieval evaluation report (2026); Multi-Task Radiotherapy Planning project report (2025). Full papers and project pages are linked through the portfolio.

AWARDS & RECOGNITION

- Generative AI & LLM Innovator - NVIDIA: Honoured for pioneering applications of Generative AI using NVIDIA's frameworks and tools.
- Innovation Award - National Science Fair: Recognized for developing a cutting-edge robotic automation model with a focus on sustainable solutions.
- Academic distinction: B.Tech First Class with Distinction; fourth-year class topper.

GOOGLE CLOUD CREDENTIALS

Silver League | 20,200 points | 15 earned badges. [Verified public Google Skills profile](#)

Implement DevOps Workflows in Google Cloud | Develop Your Google Cloud Network | Build a Secure Google Cloud Network | Set Up an App Dev Environment on Google Cloud | Implementing Cloud Load Balancing for Compute Engine | Cloud Engineering | DevOps Essentials | Kubernetes in Google Cloud